ELG 3125 - Signals & Systems

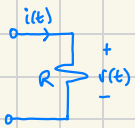
Lecture 3 - Part 2

Basic System Properties

Systems with & without memory

Memoryless: If its output for each value of indep. variable at a given time is dependent only on the input at the same time

Example: $y[n] = (2x[n] - x^2[n])^2$ → memoryless



$v(t) = Ri(t)$ → memoryless

Identity system: $y(t) = x(t)$, or $y[n] = x[n]$ → memoryless

Example of discrete-time system with memory: **Accumulator** or **Summer**

$$y[n] = \sum_{k=-\infty}^{n} x[k] = \sum_{k=-\infty}^{n-1} x[k] + x[n] = y[n-1] + x[n]$$

($\sum_{k=-\infty}^{n-1} x[k]$: 2nd last value; $x[n]$: last possible value)

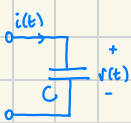
$$\Rightarrow y[n] = y[n-1] + x[n]$$

$$\Rightarrow \boxed{y[n] - y[n-1] = x[n]}$$ system with memory

$$y[n-1] = \sum_{k=-\infty}^{n-1} x[k]$$

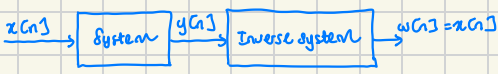
Another example: **delay** → $\boxed{y[n] = x[n-1]}$

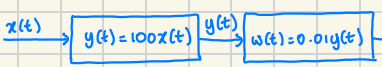
Example of continuous-time system with memory:

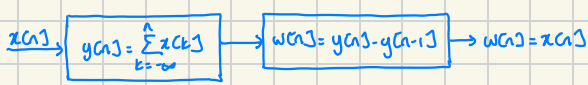


$$v(t) = \frac{1}{C}\int_{-\infty}^{t} i(t)\,dt$$

Invertibility and Inverse System



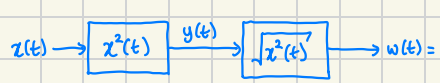




Examples of non-invertible systems:

$y[n] = 0$

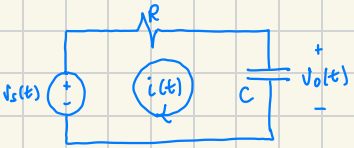
$y(t) = x^2(t)$



Causality

A system is causal if the output at any time depends on only the values of the input at present times and past times (i.e., does not anticipate future values ; nonanticipative)

Example of causal system



Capacitor C responds only to the present and past values of the source voltage.

Example of non-causal systems

$y[n] = x[n] - x[n+1]$

$y(t) = x(t+1)$

NOTE : All memoryless systems are causal, since the output responds only to the current value of input.

Example: Determine the causality of the two systems :

a) $y[n] = x[-n]$

b) $y(t) = x(t)\cos(t+1)$

a.) $n = -3 \Rightarrow y[-3] = x[-(-3)] = x[3]$ ↦ anticipates future value ⇒ not causal.

b.) Causal, the output at any time is equal to the input $x(t)$ at the same time scaled by a variational time factor $\cos(t+1)$

Stability

A stable system is one in which small inputs leads to responses that do not diverge.

If input is bounded ⇒ output also bounded ⇒ cannot diverge

if $|x(t)| < \infty \Rightarrow |y(t)| < \infty$

Check the stability of the two systems:

a) $y(t) = tx(t)$

b) $y(t) = e^{x(t)}$

a.) If $-B \leq x(t) \leq B$, it is still multiplied by t where $-\infty < t < \infty$, so $y(t)$ is not bounded.

⇒ a.) is unstable.

b) if $-B \leq x(t) \leq B \Rightarrow e^{-B} \leq y(t) \leq e^{B} \Rightarrow y(t)$ is bounded

⇒ b.) is stable

Time-Invariance

A system is **time-invariant** if a time shift in the input signal results in an identical time shift in the output signal.

If system output is $y(t)$ when input is $x(t)$ → a time-invariant system will have an output of $y(t-t_0)$ when the input is $x(t-t_0)$.

Example 1:

$y(t) = \sin[x(t)]$

Let $t = t - t_0$ → $y(t-t_0) = \sin[x(t-t_0)]$

Delay in input by t_0 → $y'(t) = \underbrace{\sin[x(t-t_0)]}_{= y(t-t_0)}$ → $\boxed{y'(t) = y(t-t_0)}$ system is time-invariant

Example 2:

$y[n] = nx[n]$

Let $n = n - n_0$ → $y[n-n_0] = (n-n_0)\,x[n-n_0]$

Delay input by n_0 → $y'[n] = nx[n-n_0] \neq y[n-n_0]$ → $\boxed{y'[n] \neq y[n-n_0]}$ system is NOT time-invariant

Example 3:

$y(t) = x(2t)$

Let: $t = t - t_0$ → $y(t-t_0) = x(2(t-t_0))$

delay input by t_0 → $y'(t) = x(2t-t_0) \neq y(t-t_0)$ → $\boxed{y'(t) \neq y(t-t_0)}$ system is NOT time-invariant

Linearity

System linear if:

- response to $x_1(t) + x_2(t)$ is $y_1(t) + y_2(t)$ - additivity property
- response to $ax_1(t)$ is $ay_1(t)$ - scaling / homogeneity property.
  - Continuous time: $ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$
  - Discrete time: $ax_1[n] + bx_2[n] \rightarrow a\,y_1[n] + by_2[n]$, $a, b \in \mathbb{C}$

Superposition property:

If $x_k[n]$, $k = 1, 2, 3, \ldots$ are inputs with corresponding outputs $y_k[n]$, $k = 1, 2, 3, \ldots$

→ response to a linear combination of these inputs is given by:

$$x[n] = \sum_k a_k x_k[n] = a_1 x_1[n] + a_2 x_2[n] + a_3 x_3[n] + \ldots$$

$$y[n] = \sum_k a_k y_k[n] = a_1 y_1[n] + a_2 y_2[n] + a_3 y_3[n] + \ldots$$

} holds for linear systems in both continuous and discrete time.

NOTE: For a linear system, zero input leads to zero output.

Example 1: $y(t) = t\,x(t)$

$y_1(t) = t\,x_1(t)$

$y_2(t) = t\,x_2(t)$

new input: $x_3(t) = a\,x_1(t) + b\,x_2(t)$

$y_3(t) = t\,x_3(t) = t[a\,x_1(t) + b\,x_2(t)] = a\underbrace{[t\,x_1(t)]}_{y_1(t)} + b\underbrace{[t\,x_2(t)]}_{y_2(t)} = a\,y_1(t) + b\,y_2(t) \Rightarrow \boxed{y_3(t) = a\,y_1(t) + b\,y_2(t)}$ system is linear

Example 2: $y(t) = x^2(t)$

$y_1(t) = x_1^2(t)$

$y_2(t) = x_2^2(t)$

New input: $x_3(t) = a\,x_1(t) + b\,x_2(t)$

$y_3(t) = x_3^2(t) = [a\,x_1(t) + b\,x_2(t)]^2 = a^2 x_1^2(t) + ab\,x_1(t)x_2(t) + ab\,x_1(t)x_2(t) + b^2 x_2^2(t) = a^2 x_1^2(t) + b^2 x_2^2(t) + 2ab\,x_1(t)x_2(t)$

$= a^2 y_1^2(t) + b^2 y_2^2(t) + 2ab\,x_1(t)x_2(t) \neq a\,y_1(t) + b\,y_2(t)$ $\Rightarrow$ Not linear